

CLASS XI · LECTURE 2

Input, Output and Storage Devices

Two periods · Printers, plotters, barcode readers, hard copy and soft copy

By the end of this lecture you can

- Classify a device as input, output, storage or communication.
- Explain a barcode reader in a superstore.
- Distinguish hard copy from soft copy, and impact printers from non-impact printers.
- Name magnetic, optical and flash storage with one example each.

Sort every device by its job

Do not memorise devices as a flat list. Place each one in a family. Input devices send data **into** the computer. Output devices present results **to** a person. Storage devices **keep** data. Communication devices **move** data between computers. The next lectures take communication and the CPU. This lecture is input, output and storage.

Input devices

Device	What it captures	Where you meet it
Keyboard	Letters, numbers and commands	Every desktop
Mouse	Position and clicks	Graphical screens
Joystick or game pad	Direction and buttons	Games and simulators
Scanner	A picture of a page	Offices
OCR	Printed characters, turned into text	Digitising a book page
OMR	Marks on a printed sheet, such as bubbles	MCQ answer sheets
MICR	Special magnetic ink characters	Cheque numbers at a bank
Barcode reader	The code printed as bars	Superstore billing
Microphone	Sound	Calls and recording
Webcam	A live picture	Online class
Touch screen	A touch, used as both input and a display	Phones and ATMs

OMR, OCR and MICR are all readers, but they do not read the same thing. **OMR** detects the position of a mark. **OCR** recognises the shape of a character. **MICR** reads characters printed in magnetic ink, which is why a bank can still read a cheque that has been stamped or a little soiled. A question that says “it is a scanning input device” and offers keyboard, mouse, trackball and OMR is asking for **OMR**.

ON THE BOARD Barcode reader in a superstore

The reader scans the bars on the packet. The code is sent to the computer. The program looks up the item name and price, adds the item to the bill, and subtracts one piece from stock. Billing is faster and the price is not typed by hand, so fewer mistakes are made.

Output devices, hard copy and soft copy**DEFINITION Hard copy and soft copy**

Hard copy is output on paper or another permanent physical medium, such as a printout. Soft copy is output on a screen. It disappears when the file is closed or the power goes off, unless you save it.

Output device	Copy	Use
Monitor (CRT, LCD, LED, plasma)	Soft	Shows text and graphics while you work
Printer	Hard	Puts the result on paper
Plotter	Hard	Draws large maps, plans and posters
Speaker or headphone	Neither paper nor screen	Sound
Multimedia projector	Soft, on a wall or screen	A class or a hall

CRT monitors are the older, deep, glass screens. LCD and LED screens are flat and use less power. LED is an LCD panel lit by light-emitting diodes. Plasma was used for large flat screens. You only need the names and the fact that all of them give soft copy.

Impact and non-impact printers

An **impact** printer strikes an inked ribbon against the paper. A dot-matrix printer does this with a column of pins. A daisy-wheel printer does it with raised characters. Impact printers are noisy and slower, but they can press through carbon paper to make several copies at once.

A **non-impact** printer never strikes the paper. An inkjet sprays ink. A laser printer uses a powder (toner) and heat. A thermal printer heats special paper. These are quiet, and the print is finer. They do not make carbon copies. A plotter is the device to name when the question says “huge drawings or maps”.

Point	Impact	Non-impact
Contact with paper	Pins or a hammer strike a ribbon	No strike
Noise	Noisy	Quiet
Carbon copies	Possible	Not in the ordinary way
Examples	Dot matrix, daisy wheel	Inkjet, laser, thermal
Typical quality	Lower, enough for bills and forms	Higher, enough for letters and photos

Storage: where data lives

Primary memory (RAM and ROM) sits on the motherboard and is covered in the next lecture. Secondary storage keeps data when the power is off. Group it by the way it records.

Family	How it records	Examples	Exam point
Magnetic disk	Magnetised spots on a spinning surface	Hard disk, and the older floppy disk	Floppy disk and hard disk are magnetic disks, not optical disks
Optical disc	Pits read by a laser	CD, DVD, Blu-ray	Good for software, films and backups
Flash / solid state	Electronic cells, no spinning parts	USB pen drive, SD card, SSD	Small, fast, shock-resistant
Magnetic tape	A long strip, read in order	Backup cartridges	Cheap for archives, slow if you need one file in the middle

A **SIMM** is a Single Inline Memory Module: a small circuit board of RAM chips that plugs into the motherboard. It is a package for main memory, not a secondary-storage disk. The name is a favourite one-mark question. A later package, DIMM, means Dual Inline Memory Module.

EXAM TIP Four contrasts that return every year

Hard copy / soft copy. Impact / non-impact. Magnetic disk / optical disc. OMR / OCR / MICR. Write the contrast in a two-column form even in a short answer. Markers look for the difference, not for a paragraph that describes only one side.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. Floppy disk and hard disk are types of:

- A) Optical disc
- B) Magnetic tape
- C) Magnetic disk
- D) Flash memory

2. SIMM stands for:

- A) Simple Inline Memory Mode
- B) Single Inline Memory Module
- C) Serial Interface Memory Module
- D) Single Internal Memory Map

3. Which of these is a scanning input device?

- A) Keyboard
- B) Mouse
- C) Trackball
- D) OMR

Short answers

S1. How is a barcode reader useful in a superstore?

S2. Write four differences between impact and non-impact printers.

S3. Differentiate hard copy and soft copy.

Detailed answer

L1. Explain secondary storage devices with examples.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. C. Both record data as magnetic spots on a disk.

MCQ 2. B. It is a small board of RAM chips.

MCQ 3. D. OMR scans marks. The others point or type.

S1. A barcode reader scans the bars printed on a product and sends that code to the computer. The program finds the item name and price, adds the item to the customer's bill, and updates the stock. Billing is quicker and the cashier does not type the price, so errors are fewer.

S2. An impact printer strikes a ribbon (dot matrix); a non-impact printer does not (laser or inkjet). Impact printers are noisy; non-impact printers are quiet. Impact printers can make carbon copies; non-impact printers normally cannot. Non-impact printers give finer print and are preferred for letters and pictures.

S3. Hard copy is output printed on paper, such as a mark sheet from a printer. It can be filed and read without a computer. Soft copy is output shown on a monitor. It is easy to change and needs no paper, but it is gone from the screen when the file is closed unless it has been saved.

L1. Secondary storage keeps data and programs when the power is off, and it holds much more than RAM. Magnetic disks, such as the hard disk and the older floppy disk, store data as magnetised spots; the hard disk is the usual permanent store inside a system unit. Optical discs, such as CD, DVD and Blu-ray, are read by a laser and are used for software, films and backups. Flash storage, such as a USB pen drive, an SD card or an SSD, stores data in electronic cells, has no spinning parts and is fast and portable. Magnetic tape stores data in order along a strip and is used for large backups, though reaching one file in the middle is slow. A plotter is not storage: it is an output device for large drawings.